Luan Thach Hoang

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Webpage: `https://www.math.ttu.edu/~lhoang/`

Education

- 2005 — *Ph.D. in Mathematics*
  Texas A&M University, College Station, Texas
- 2000 — *M.A. in Mathematics*
  Arizona State University, Tempe, Arizona
- 1997 — *B.S. in Mathematics*
  National University, Hochiminh City, Vietnam
- 1997 — *B.S. in Information Technology*
  National University, Hochiminh City, Vietnam

Employment

- 9.2014 – present — Texas Tech University, Lubbock, Texas
  Associate professor
- 9.2008 – 8.2014 — Texas Tech University, Lubbock, Texas
  Assistant professor
- 9.2005 – 5.2008 — University of Minnesota, Minneapolis, Minnesota
  Dunham Jackson assistant professor
- 6.2004 – 12.2004 — Texas A&M University, College Station, Texas
  Teaching assistant
- 9.2000 – 8.2002 — Indiana University, Bloomington, Indiana
  Associate instructor
- 9.1998 – 8.2000 — Arizona State University, Tempe, Arizona
  Teaching assistant, research assistant
- 9.1997 – 8.1998 — National University, Hochiminh City, Vietnam
  Instructor

Research Interest

Partial differential equations, fluid dynamics, porous media equations, dynamical systems.

Publications

55. Luan Hoang, Akif Ibragimov, *Linear non-divergence parabolic equations in non-cylindrical space-time sets,* 1–58, submitted.
    DOI: `10.48550/arXiv.2608.02221`

54. Emine Celik, Luan Hoang, Thinh Kieu, *A priori estimates for gaseous flows of Forchheimer-type in heterogeneous porous media,* 1–43, submitted.
    DOI: `10.48550/arXiv.2505.11987` (Preprint)

53. Luan Hoang, *Asymptotic expansions with subordinate variables for solutions of the Navier–Stokes equations,* 1–40, submitted.
DOI: `10.48550/arXiv.2403.03132` (Preprint)

52. Luan Hoang, Michael S. Jolly, (with an appendix by Chengzhang Fu,) *An intrinsic expansion approach to the Galerkin approximations for the Navier–Stokes equations,* Discrete and Continuous Dynamical Systems, 1–29, accepted.
DOI: `10.48550/arXiv.2602.16064` (Preprint)

51. Luan Hoang, Akif Ibragimov, *Linear non-divergence elliptic equations in a bounded, infinitely winding planar domain,* Complex Variables and Elliptic Equations (2026), 1–31, in press.
DOI: `10.1080/17476933.2026.2630197`

50. Emine Celik, Luan Hoang, Thinh Kieu, *On compressible fluid flows of Forchheimer-type in rotating heterogeneous porous media,* Mathematical Methods in the Applied Sciences (2026), Vol. 49, No. 13, 15010–15055.
DOI: `10.1002/mma.70807`

49. Luan Hoang, *A new form of asymptotic expansion for non-smooth differential equations with time-decaying forcing functions,* Differential Equations and Dynamical Systems, Vol. 34, No. 3, 697–741, 2026.
DOI: `10.1007/s12591-024-00703-z`

48. Luan Hoang, Akif Ibragimov, *Asymptotic estimates for solutions of inhomogeneous non-divergence diffusion equations with drifts,* Journal of Mathematical Analysis and Applications, Volume 560, Issue 2, 15 August 2026, 130489 (34 pages).
DOI: `10.1016/j.jmaa.2026.130489`

47. Luan Hoang, *On the finite time blow-ups for solutions of nonlinear differential equations,* Evolution Equations and Control Theory (June 2026), Vol. 20, 91–121.
DOI: `10.3934/eect.2026022`

46. Luan Hoang, Akif Ibragimov, *A class of anisotropic diffusion-transport equations in non-divergence form,* Contemporary Mathematics. Fundamental Directions (2025), Vol. 71, No. 4, 663–685. (Russian) The English version is to appear in Journal of Mathematical Sciences.
DOI: `10.22363/2413-3639-2025-71-4-663-685` (Russian)
DOI: `10.48550/arXiv.2503.03089` (English Preprint)

45. Luan Hoang, *Behavior near the extinction time for systems of differential equations with sublinear dissipation terms,* Electronic Journal of Differential Equations, Vol. 2025 (2025), No. 08, 1–26.
DOI: `10.58997/ejde.2025.08`

44. Luan Hoang, Thinh Kieu, *Anisotropic flows of Forchheimer-type in porous media and their steady states,* Nonlinear Analysis: Real World Applications (August 2025), Vol. 84, 104269 (30 pp).
DOI: `10.1016/j.nonrwa.2024.104269`

43. Luan Hoang, Michael S. Jolly, *Intrinsic expansions in large Grashof numbers for the steady states of the Navier–Stokes equations,* Nonlineary (2024), Vol. 37, No. 12, 125008 (32 pp).
DOI: `10.1088/1361-6544/ad84aa`

42. Luan Hoang, *Asymptotic expansions about infinity for solutions of nonlinear differential equations with coherently decaying forcing functions,* Annali della Scuola Normale Superiore di Pisa, Classe di Scienze, Vol. XXV, No. 1 (2024), 311–370.
DOI: `10.2422/2036-2145.202109_004`

41. Ciprian Foias, Luan Hoang, Michael S. Jolly, *On Galerkin approximations of the Navier-Stokes equations in the limit of large Grashof numbers,* Communications on Pure and Applied Analysis, Volume 23, Issue 2 (2024), 269–303.
DOI: `10.3934/cpaa.2024010`

40. Luan Hoang, *The Navier–Stokes equations with body forces decaying coherently in time,* Journal of Mathematical Analysis and Applications, Vol. 531, Issue 2, Part 1 (2024), 127863 (39 pp).
DOI: `10.1016/j.jmaa.2023.127863`

39. Luan Hoang, *Long-time behaviour of solutions of superlinear systems of differential equations,* Dynamical Systems, Vol. 39, No. 1 (2024), 79–107.
DOI: `10.1080/14689367.2023.2234845`

38. Emine Celik, Luan Hoang, Thinh Kieu, *Studying a doubly nonlinear model of slightly compressible Forchheimer flows in rotating porous media,* Turkish Journal of Mathematics, Turkish Journal of Mathematics, Vol. 47, No. 3 (2023), 949–987.
DOI: `10.55730/1300-0098.3405`

37. Dat Cao, Luan Hoang, Thinh Kieu, *Infinite series asymptotic expansions for decaying solutions of dissipative differential equations with non-smooth nonlinearity,* Qualitative Theory of Dynamical Systems, Volume 20, Issue 3 (2021), 62, 38 pp.
DOI: `10.1007/s12346-021-00502-9`

36. Emine Celik, Luan Hoang, Thinh Kieu, *Slightly compressible Forchheimer flows in rotating porous media,* Journal of Mathematical Physics, Volume 62 (2021), Issue 7, 073101, 39 pp.
DOI: `10.1063/5.0047754`

35. Dat Cao, Luan Hoang, *Asymptotic expansions with exponential, power, and logarithmic functions for non-autonomous nonlinear differential equations,* Journal of Evolution Equations, Volume 21, Issue 2, 1179–1225, 2021.
DOI: `10.1007/s00028-020-00622-w`

34. Luan Hoang, *Asymptotic expansions for the Lagrangian trajectories from solutions of the Navier–Stokes equations,* Communications in Mathematical Physics, Volume 383, Issue 2, 981–995 (2021).
DOI: `10.1007/s00220-020-03863-5`

33. Luan Hoang, Edriss Titi, *Asymptotic expansions in time for rotating incompressible viscous fluids,* Annales de l'Institut Henri Poincaré. Analyse Non Linéaire, Volume 38, Issue 1, January–February 2021, 109–137.
DOI: `10.1016/j.anihpc.2020.06.005`

32. Dat Cao, Luan Hoang, *Asymptotic expansions in a general system of decaying functions for solutions of the Navier-Stokes equations,* Annali di Matematica Pura ed Applicata, Vol. 199, No. 3 (2020), 1023–1072.
DOI: `10.1007/s10231-019-00911-3`

31. Dat Cao, Luan Hoang, *Long-time asymptotic expansions for Navier–Stokes equations with power-decaying forces,* Proceedings of the Royal Society of Edinburgh: Section A Mathematics, Vol. 150, No. 2 (2020), 569–606.
DOI: `10.1017/prm.2018.154`

30. Luan Hoang, Thinh Kieu, *Global estimates for generalized Forchheimer flows of slightly compressible fluids,* Journal d'Analyse Mathematique, March 2019, Volume 137, Issue 1, 1–55.
DOI: `10.1007/s11854-018-0064-5`

29. Emine Celik, Luan Hoang, Thinh Kieu, *Doubly nonlinear parabolic equations for a general class of Forchheimer gas flows in porous media,* Nonlinearity, Vol. 31, No. 8 (2018) 3617–3650.
DOI: `10.1088/1361-6544/aabf05`

28. Ciprian Foias, Luan Hoang, Jean-Claude Saut, *Navier and Stokes meet Poincaré and Dulac,* J. Appl. Anal. Comput., Volume 8, Number 3, (June 2018) 727–763. (survey)
DOI: `10.11948/2018.727`

27. Luan Hoang, Vincent Martinez, *Asymptotic expansion for solutions of the Navier-Stokes equations with non-potential body forces,* J. Math. Anal. Appl., Volume 462, Issue 1, (1 June 2018) 84–113.
DOI: `10.1016/j.jmaa.2018.01.065`

26. Emine Celik, Luan Hoang, Thinh Kieu, *Generalized Forchheimer flows of isentropic gases,* J. Math. Fluid Mech., (March 2018) Volume 20, Issue 1, 83–115.
DOI: `10.1007/s00021-016-0313-2`

25. Luan Hoang, Eric Olson, James Robinson, *Continuity of pullback and uniform attractors,* J. Differential Equations, Volume 264, Issue 6, (15 March 2018) 4067–4093.
DOI: `10.1016/j.jde.2017.12.002`

24. Luan Hoang, Thinh Kieu, *Interior estimates for generalized Forchheimer flows of slightly compressible fluids,* Advanced Nonlinear Studies, 17(4), (October 2017) 739–767.
DOI: `10.1515/ans-2016-6027`

23. Luan Hoang, Vincent Martinez, *Asymptotic expansion in Gevrey spaces for solutions of Navier-Stokes equations,* Asymptotic Analysis, (2017), vol. 104, no. 3–4, 167–190.
DOI: `10.3233/ASY-171429`

22. Emine Celik, Luan Hoang, Akif Ibragimov, Thinh Kieu, *Fluid flows of mixed regimes in porous media,* J. Math. Phys., Volume 58 (2017), No. 2, 023102, 30 pp.
DOI: `10.1063/1.4976195`

21. Emine Celik, Luan Hoang, *Maximum estimates for generalized Forchheimer flows in heterogeneous porous media,* J. Differential Equations, Volume 262, Issue 3 (5 February 2017), 2158–2195.
DOI: `10.1016/j.jde.2016.10.043`

20. Luan Hoang, Truyen Nguyen, Tuoc Phan, *Local gradient estimates for degenerate elliptic equations,* Advanced Nonlinear Studies, Volume 16, Issue 3 (Aug 2016), 479–489.
DOI: `10.1515/ans-2015-5038`

19. Emine Celik, Luan Hoang, *Generalized Forchheimer flows in heterogeneous porous media,* Nonlinearity, Volume 29, Number 3 (March 2016), 1124–1155.
DOI: `10.1088/0951-7715/29/3/1124`

18. Luan Hoang, Akif Ibragimov, Thinh Kieu, Zeev Sobol, *Stability of solutions to generalized Forchheimer equations of any degree,* Journal of Mathematical Sciences (via journal "Problems in Mathematical Analysis"), Volume 210, Number 4 (2015), 476–544.
DOI: `10.1007/s10958-015-2576-1`

17. Luan Hoang, Eric Olson, James Robinson, *On the continuity of global attractors,* Proc. Amer. Math. Soc., Volume 143, Number 10 (2015), 4389–4395.
DOI: `10.1090/proc/12598`

16. Luan Hoang, Truyen Nguyen, Tuoc Phan, *Gradient estimates and global existence of smooth solutions to a cross-diffusion system,* SIAM Journal on Mathematical Analysis, SIAM Journal on Mathematical Analysis, Volume 47, Issue 3 (2015), 2122–2177.
DOI: `10.1137/140981447`

15. Luan Hoang, Akif Ibragimov, Thinh Kieu, *A family of steady two-phase generalized Forchheimer flows and their linear stability analysis,* J. Math. Phys. 55, Issue 12 (2014), 123101, 32 pp.
DOI: `10.1063/1.4903002`

14. Luan Hoang, Thinh Kieu, Tuoc Phan, *Properties of generalized Forchheimer flows in porous media,* "Problems of Mathematical Analysis", volume 76 (July-August 2014), and in "Journal of Mathematical Sciences", Vol. 202 No. 2 (October 2014), 259–332.
DOI: `10.1007/s10958-014-2045-2`

13. Luan Hoang, *Incompressible fluids in thin domains with Navier friction boundary conditions (II),* Journal of Mathematical Fluid Mechanics, Volume 15, Issue 2, (June 2013) 361–395. DOI: `10.1007/s00021-012-0123-0`

12. Luan Hoang, Akif Ibragimov, Thinh Kieu, *One-dimensional two-phase generalized Forchheimer flows of incompressible fluids,* J. Math. Anal. Appl., Volume 401, Issue 2, (May 2013) 921–938. DOI: `10.1016/j.jmaa.2012.12.055`

11. Luan Hoang, Akif Ibragimov, *Qualitative study of generalized Forchheimer flows with the flux boundary condition,* Advances in Differential Equations, Volume 17, Numbers 5-6, (May/June 2012) 511–556. DOI: `10.57262/ade/1355703078`

10. Ciprian Foias, Luan Hoang, Jean-Claude Saut, *Asymptotic integration of Navier-Stokes equations with potential forces. II. An explicit Poincaré-Dulac normal form*, Journal of Functional Analysis, Vol. 260, Issue 10 (2011), 3007–3035. DOI: `10.1016/j.jfa.2011.02.005`

9. Luan Hoang, Akif Ibragimov, *Structural stability of generalized Forchheimer equations for compressible fluids in porous media*, Nonlinearity, Volume 24, Number 1 (January 2011) 1–41. DOI: `10.1088/0951-7715/24/1/001`

8. Luan Hoang, George R Sell, *Navier–Stokes equations with Navier boundary conditions for an oceanic model*, Journal of Dynamics and Differential Equations, Volume 22, Number 3 (September 2010), 563–616. DOI: `10.1007/s10884-010-9189-7`

7. Luan Hoang, *Incompressible fluids with Navier friction boundary conditions in thin domains (I)*, Journal of Mathematical Fluid Mechanics, Volume 12, Number 3 (August 2010), 435–472. DOI: `10.1007/s00021-009-0297-2`

6. Eugenio Aulisa, Lidia Bloshanskaya, Luan Hoang, Akif Ibragimov, *Analysis of generalized Forchheimer equations of compressible fluids in porous media*, Journal of Mathematical Physics 50, Issue 10, (2009), 103102, 44 pp. DOI: `10.1063/1.3204977`

5. Ciprian Foias, Luan Hoang, Basil Nicolaenko, *On the helicity in 3D Navier–Stokes equations II: The statistical case*, Communications in Mathematical Physics, Volume 290, Issue 2 (2009), 679–717. DOI: `10.1007/s00220-009-0827-z`

4. Ciprian Foias, Luan Hoang, Eric Olson, Mohammed Ziane, *The normal form of the Navier–Stokes equations in suitable normed spaces*, Annales de l'Institute Henri Poincaré - Analyse Non Linéaire, Volume 26, Issue 5 (September-October 2009), 1635–1673. DOI: `10.1016/j.anihpc.2008.09.003`

3. Luan Hoang, *A basic inequality for the Stokes operator related to the Navier boundary condition*, Journal of Differential Equations, Volume 245, Issue 9 (November 2008), 2585–2594. DOI: `10.1016/j.jde.2008.01.024`

2. Ciprian Foias, Luan Hoang, Basil Nicolaenko, *On the helicity in 3D Navier–Stokes equations I: The non-statistical case*, Proceedings of the London Mathematical Society, Volume 94 Part 1 (January 2007) 53–90. DOI: `10.1112/plms/pdl003`

1. Ciprian Foias, Luan Hoang, Eric Olson, Mohammed Ziane, *On the solutions to the normal form of the Navier–Stokes equations*, Indiana University Mathematics Journal, Vol. 55, No 2 (2006) 631–686. DOI: `10.1512/iumj.2006.55.2830`

Funding

- 2014–2017 Title: *Nonlinear Couplings for Flows in Fractured Porous Media: Analysis and Numerical Algorithms*
  Agency: NSF - Applied Mathematics
  Amount: $290,001.00
  Role: Co-principal investigator
  Status: **Funded, DMS 1412796**

- 2009–2012 Title: *Analysis of non-linear flows in heterogeneous porous media and applications*
  Agency: NSF - Applied Mathematics
  Amount: $221,626
  Role: Co-principal investigator
  Status: **Funded, DMS 0908177**

- 2009 Title: *Mini-Symposium on Nonlinear Analysis, PDE, and Applications*
  Agency: NSF - Applied Mathematics
  Amount: $15,000
  Role: Principal investigator
  Status: **Funded, DMS 0931596**

Conferences

- 3.29.2025 AMS 2025 Spring Central Sectional Meeting
  University of Kansas, Lawrence, Kansas, March 29–30, 2025
  Invited Talk: *Intrinsic expansions in large Grashof numbers for the steady states of the Navier-Stokes equations*

- 3.8.2025 The 2025 Texas Differential Equations Conference
  University of Houston-Downtown, March 8, 2025
  Invited Talk: *Anisotropic Forchheimer flows in porous media*

- 10.13.2024 SIAM Conference Texas-Louisiana Section, 7th Annual Meeting
  Baylor University, Waco, Texas, October 11–13, 2024
  Invited Talk: *On Galerkin approximations of the Navier-Stokes equations with large Grashof numbers*

- 7.12.2024 The 2024 SIAM Annual Meeting
  Spokane, Washington, July 8–12, 2024
  Invited Talk: *Asymptotic expansions for the Navier–Stokes equations*

- 2.21.2022 Workshop "Quasi-linear PDEs in Fluid II"
  Virtual, jointly organized by Japan, Korea, U.S., February 20 – 21, 2022
  Invited Talk: *Open problems in asymptotic expansions for viscous incompressible fluids*

- 9.18.2021 The 44th SIAM Southeastern Atlantic Section Conference
  Auburn University, Auburn, Alabama, September 18-19, 2021
  Invited Talk: *Infinite series asymptotic expansions for solutions of dissipative nonlinear differential equations*

- 7.25.2021 Saigon Summer Meeting 2021
  Online, July 24-25, 2021
  Invited Talk: *Large time asymptotic behaviors of fluids in the Eulerian and Lagrangian formulations*

- 1.7.2021 — 2021 Joint AMS-MAA Mathematics Meeting
  Virtual, January 6-9, 2021
  Invited Talk: *Asymptotic expansions for decaying solutions of dissipative differential equations*

- 9.12.2020 — AMS 2020 Fall Central Sectional Meeting
  Virtual, September 12-13, 2020.
  Invited Talk: *Asymptotic expansions for the Lagrangian trajectories from solutions of the Navier-Stokes equations*

- 10.12.2019 — AMS 2019 Fall Eastern Sectional Meeting
  Binghamton University, Binghamton, New York, October 12-13, 2019
  Invited Talk: *Asymptotic expansions for rotating incompressible viscous fluids*

- 10.6.2018 — The 4th Annual Meeting of SIAM Central States Section
  Norman, Oklahoma, October 5-7, 2018
  Invited Talk: *Developments in asymptotic expansions for solutions of Navier-Stokes equations*

- 1.4.2017 — 2017 Joint Mathematics Meeting
  Atlanta, Georgia, January 4-7, 2017
  Invited Talk: *Asymptotic expansion for solutions of Navier-Stokes equations with a non-potential body force*

- 7.3.2016 — The 11th AIMS Conference on Dynamical Systems, Differential Equations and Applications
  Orlando, Florida, July 1-5, 2016
  Invited Talk: *Global existence of smooth solutions to the SKT system in high dimensional spaces*

- 5.18.2016 — International Conference on Evolution Equations in conjunction with the 31st annual Shanks Lecture
  Vanderbilt University, Nashville, TN, May 16-20, 2016
  Invited Talk: *Asymptotic expansion for solutions of Navier-Stokes equations with a non-potential body force*

- 5.16.2016 — 46th Annual John H. Barrett Memorial Lectures
  University of Tennessee, Knoxville, TN, May 16-18, 2016
  Invited Talk: *Regular solutions of the SKT system in any dimensions*

- 3.12.2016 — The 40th SIAM Southeastern Atlantic Section Conference (SIAM-SEAS). Applied Mathematics
  University of Georgia, Athens, Georgia, March 12-13, 2016
  Invited Talk: *Asymptotic expansion for solutions of Navier-Stokes equations*

- 1.8.2016 — 2016 Joint Mathematics Meeting
  Seattle, Washington, January 6-9, 2016
  Invited Talk: *On non-Darcy fluid flows in porous media*

- 12.7.2015 — SIAM Conference on Analysis of Partial Differential Equations
  Scottsdale, Arizona, December 7-10, 2015
  Talk: *Continuity of attractors for dynamical systems*

- 4.19.2015 — AMS 2015 Spring Western Sectional Meeting
  Las Vegas, NV, April 18-19, 2015
  Invited Talk: *On the normal form of Navier-Stokes equations in Gevrey spaces*

- 7.10.2014 — The 10th AIMS Conference on Dynamical Systems, Differential Equations and Applications
  Madrid, Spain, July 7-11, 2014
  Invited Talk: *Estimates in $W^{1,\infty}$ for generalized Forchheimer equations in porous media*

- 7.1.2014 — Advances in Mathematical Fluid Mechanics, Stochastic and Deterministic Methods
  Lisbon, Portugal, June 30-July 5, 2014
  Invited Talk: *On two-phase Forchheimer flows of incompressible fluids*

- 4.12.2014 — AMS 2014 Spring Central Sectional Meeting
  Texas Tech University, Lubbock, TX, April 11-13, 2014
  Invited Talk: *Derivative estimates for generalized Forchheimer flows*

- 12.9.2013 — SIAM conference on Analysis of Partial Differential Equations
  Lake Buena Vista, Florida, December 7-10, 2013
  Invited Talk: *On dynamics of fluid flows in porous media*

- 10.28.2012 — AMS 2012 Fall Western Section Meeting
  University of Arizona, Tucson, AZ, October 27-28, 2012
  Invited Talk: *The Stokes operator for an interface boundary value problem in two-layer domains*

- 10.21.2012 — AMS 2012 Central Fall Section Meeting
  University of Akron, Akron, OH, October 20-21, 2012
  Invited Talk: *Generalized Forchheimer equations for slightly compressible fluids*

- 7.4.2012 — The 9th AIMS Conference on Dynamical Systems, Differential Equations and Applications
  Orlando, Florida, July 1-5, 2012
  Invited Talk: *A Poincaré–Dulac normal form for Navier–Stokes equations*

- 11.17.2011 — SIAM Conference on Analysis of Partial Differential Equations
  San Diego, CA, November 14-17, 2011
  Invited Talk: *Navier–Stokes equations in thin two-layer domains with non-flat boundaries*

- 11.16.2011 — SIAM Conference on Analysis of Partial Differential Equations
  San Diego, CA, November 14-17, 2011
  Invited Talk: *Analysis of non-Darcy compressible flows in porous media*

- 3.26.2011 — 34th Annual Texas Differential Equations Conference
  University of Texas-Pan American, Edinburg, TX, March 26-27, 2011
  Invited Talk: *Dynamics and Stabilities of Generalized Forchheimer Flows in Porous Media*

- 10.10.2010 — AMS 2010 Fall Western Section Meeting
  Los Angeles, CA, October 9-10, 2010
  Invited Talk: *Structural stability of nonlinear flows in porous media*

- 4.25.2009 — AMS 2009 Spring Western Section Meeting
  San Francisco, CA, April 25-26, 2009
  Invited Talk: *Generalized Forchheimer equations in porous media*

- 5.19.2008 — The 7th AIMS International Conference on Dynamical Systems and Differential Equations
  Arlington, TX, May 18-21, 2008
  INVITED TALK: *On the normal form of the Navier–Stokes equations in suitable Banach spaces*

- 12.11.2007 — SIAM Conference on Analysis of Partial Differential Equations
  Mesa, AZ, December 10-12, 2007
  INVITED TALK: *Incompressible fluids in thin domains with Navier friction boundary conditions*

- 11.19.2007 — Nonlinear Dynamics and PDE Mini-Conference
  Arizona State University, Tempe, AZ, November 19-20, 2007
  INVITED TALK: *Navier–Stokes equations: the normalization map, statistical solutions and fluid dynamics*

- 11.4.2007 — AMS 2007 Fall Southeastern Meeting
  Murfreesboro, TN, November 3-4, 2007
  INVITED TALK: *Global strong solutions of equations in geophysical fluid dynamics*

- 5.28.2007 — The 3rd Symposium on Analysis & PDEs
  Purdue University, West Lafayette, IN, May 27-30, 2007
  CONTRIBUTED TALK: *Regularity of the Stokes operator in thin domains*

- 5.21.2007 — US–Chile Workshop on New Developments in Partial Differential Equations I
  Carnegie Mellon University, Pittsburgh, PA, May 21-24, 2007
  CONTRIBUTED TALK: *Navier–Stokes equations with Navier boundary conditions in nearly flat domains*

- 3.17.2007 — AMS 2007 Spring Central Section Meeting
  Oxford, OH, March 16-17, 2007
  INVITED TALK: *Statistical solutions to the Navier–Stokes equations and long time behaviors of fluid flows*

- 3.4.2007 — AMS 2007 Spring Southeastern Section Meeting
  Davidson, NC, March 3-4, 2007
  INVITED TALK: *Studying the normal form of the Navier–Stokes equations in suitable Banach spaces*

- 3.18.2005 — AMS 2005 Spring Southeastern Sectional Meeting
  Bowling Green, KY, March 18-19
  INVITED TALK: *On the solutions to the normal form of the Navier–Stokes Equations*

- 12.2004 — SIAM Conference on Analysis of Partial Differential Equations
  Houston, TX, December 6-8, 2004
  INVITED TALKS: *On the convergence of the asymptotic expansions of the regular solutions to the 3D-periodic Navier–Stokes equations and applications to asymptotic behavior of helicity.* Parts I and II.

- 4.3.2004 — AMS 2004 Spring Western Section Meeting
  Los Angeles, CA, April 3-4
  INVITED TALK: *On the helicity in 3D Navier–Stokes equations*

Seminars and Colloquia

- 12.1.2025 V. I. Smirnov Seminar on Mathematical Physics
St. Petersburg Department of Steklov Mathematical Institute
Invited Talk: *A class of anisotropic diffusion-transport equations in non-divergence form*

- 9.15.2025 Analsyis Seminar
Department of Mathematics and Statistics, Texas Tech University
Invited Talk: *On the finite time blow-ups for solutions of nonlinear differential equations. Part II*

- 9.8.2025 Analsyis Seminar
Department of Mathematics and Statistics, Texas Tech University
Invited Talk: *On the finite time blow-ups for solutions of nonlinear differential equations. Part I*

- 3.28.2025 Applied Mathematics Seminar
Department of Mathematics, Kansas State University
Invited Talk: *On the Forchheimer flows in anisotropic porous media*

- 2.3.2025 Analsyis Seminar
Department of Mathematics and Statistics, Texas Tech University
Invited Talk: *Asymptotic expansions with a subordinate variable for non-smooth differential equations. Part II*

- 1.27.2025 Analsyis Seminar
Department of Mathematics and Statistics, Texas Tech University
Invited Talk: *Asymptotic expansions with a subordinate variable for non-smooth differential equations. Part I*

- 9.23.2024 Analsyis Seminar
Department of Mathematics and Statistics, Texas Tech University
Invited Talk: *Anisotropic flows of Forchheimer-type in porous media. Part II*

- 9.16.2024 Analsyis Seminar
Department of Mathematics and Statistics, Texas Tech University
Invited Talk: *Anisotropic flows of Forchheimer-type in porous media. Part I*

- 2.19.2024 Analsyis Seminar
Department of Mathematics and Statistics, Texas Tech University
Invited Talk: *On Galerkin approximations of the Navier–Stokes equations in the limit of large Grashof numbers (Part II)*

- 2.12.2024 Analsyis Seminar
Department of Mathematics and Statistics, Texas Tech University
Invited Talk: *On Galerkin approximations of the Navier–Stokes equations in the limit of large Grashof numbers*

- 11.13.2023 V. I. Smirnov Seminar on Mathematical Physics
St. Petersburg Department of Steklov Mathematical Institute
Invited Talk: *Complicated asymptotic expansions for the Navier–Stokes equations*

- 9.18.2023 Analsyis Seminar
Department of Mathematics and Statistics, Texas Tech University
Invited Talk: *On the finite time extinction for nonlinear differential equations (Part II)*

- 9.11.2023 — Analsyis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *On the finite time extinction for nonlinear differential equations (Part I)*

- 4.5.2023 — SIAM Student Chapter Meeting
  Department of Mathematics, University of North Georgia
  Invited Talk: *Behavior near the extinction time for systems of differential equations with sublinear dissipation terms*

- 3.6.2023 — PDE Seminar
  Department of Mathematics, Indiana University
  Invited Talk: *Asymptotic expansions for solutions of the Navier–Stokes equations with body forces decaying coherently in time*

- 1.23.2023 — Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Long-time behavior of solutions of superlinear systems of differential equations*

- 9.19.2022 — Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *The Navier–Stokes equations with body forces decaying coherently in time*

- 3.21.2022 — Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *A doubly nonlinear model of slightly compressible Forchheimer flows in rotating porous media*

- 10.4.2021 — Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Asymptotic expansions about infinity for solutions of nonlinear differential equations with coherently decaying forcing functions*

- 9.9.2021 — PDE Seminar
  Department of Mathematics
  University of Tennessee, Knoxville
  Invited Talk: *Asymptotic analysis for viscous, incompressible fluids*

- 2.22.2021 — Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Infinite series asymptotic expansions for dissipative differential equations with non-smooth nonlinearity*

- 1.26.2021 — Nonlinear PDEs Seminar,
  Department of Mathematics, Texas A&M University
  Invited Talk: *The Navier-Stokes equations: asymptotic expansions for solutions and their associated Lagrangian trajectories*

- 10.8.2020 — Applied Mathematics Seminar
  Department of Mathematics and Statistics, Hunter College
  Invited Talk: *Long-time asymptotic expansions for viscous incompressible fluid flows*

- 9.21.2020 — Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Asymptotic analysis of the Lagrangian trajectories from solutions of the Navier-Stokes equations*

- 3.9.2020 Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Asymptotic expansions for solutions of the Navier–Stokes–Boussinesq equations. Parts II*

- 3.2.2020 Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Asymptotic expansions for solutions of the Navier–Stokes–Boussinesq equations. Parts I*

- 10.7.2019 Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Slightly Compressible Forchheimer Flows in Rotating Porous Media. Parts II*

- 9.30.2019 Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Slightly Compressible Forchheimer Flows in Rotating Porous Media. Parts I*

- 4.15.2019 Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Asymptotic expansions for decaying solutions of ODEs. Parts II*

- 4.8.2019 Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Asymptotic expansions for decaying solutions of ODEs. Parts I*

- 11.26.2018 Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Asymptotic expansions in time for solutions of Navier-Stokes equations of rotating fluids*

- 9.27.2018 Colloquium
  Department of Mathematics and Statistics, Texas Tech University
  Talk: *Analysis of Navier-Stokes systems and Forchheimer flows*

- 3.5.2018 Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Asymptotic expansions in Gevrey spaces for solutions of Navier-Stokes equations in periodic domains*

- 2.26.2018 Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Gevrey classes and the Navier-Stokes equations. Part II*

- 12.4.2017 Differential Equations Seminar
  Department of Mathematics and Statistics, University of Maryland Baltimore County
  Invited Talk: *Studying nonlinear fluid flows in heterogeneous porous media*

- 11.20.2017 Analysis Seminar
  Department of Mathematics and Statistics, Texas Tech University
  Invited Talk: *Gevrey classes and the Navier-Stokes equations*

- 11.10.2017 Joint PDEs and Mathematical Physics Seminar,
Department of Mathematics, Texas A&M University
INVITED TALK: *Large-time asymptotic expansions for solutions of Navier-Stokes equations*

- 11.3.2017 Colloquium
Department of Mathematics, University of North Georgia
INVITED TALK: *Asymptotic expansions in large time for solutions of non-autonomous differential equations*

- 10.16.2017 Differential Equations and Applied Math Seminar,
Department of Mathematics, University of Louisville
INVITED TALK: *Foias-Saut expansions for solutions of nonlinear differential equations*

- 9.27.2017 Seminar
Institute for Scientific Computing and Applied Mathematics, Indiana University
INVITED TALK: *Asymptotic expansions of Foias-Saut type for Navier-Stokes equations with decaying non-potential forces*

- 9.14.2017 Differential Equations Seminar
Department of Mathematics, University of Tennessee
INVITED TALK: *Foias-Saut asymptotic expansions for solutions of Navier-Stokes equations with time-dependent forces*

- 3.20.2017 Analysis Seminars
Department of Mathematics and Statistics, Texas Tech University
INVITED TALK: *Models and analysis of fluid flows in heterogeneous porous media (Part II)*

- 3.6.2017 Analysis Seminars
Department of Mathematics and Statistics, Texas Tech University
INVITED TALK: *Models and analysis of fluid flows in heterogeneous porous media (Part II)*

- 12.2.2016 Colloquium
Department of Mathematics and Statistics, University of Maryland, Baltimore County
INVITED TALK: *On the theory of asymptotic expansions and normal form for Navier-Stokes equations*

- 2.17.2016 Applied Mathematics Seminars
Department of Mathematics and Statistics, Texas Tech University
TALK: *Continuity of global, pullback and uniform attractors*

- 9.16.2015 Applied Mathematics Seminars
Department of Mathematics and Statistics, Texas Tech University
TALK: *Global estimates for generalized Forchheimer flows of slightly compressible fluids. Part II.*

- 9.9.2015 Applied Mathematics Seminars
Department of Mathematics and Statistics, Texas Tech University
TALK: *Global estimates for generalized Forchheimer flows of slightly compressible fluids*

- 9.17.2014 Applied Mathematics Seminars
Department of Mathematics and Statistics, Texas Tech University
TALK: *Analysis of single and multi phase flows in porous media (III)*

- 9.10.2014 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  Talk: *Analysis of single and multi phase flows in porous media (II)*

- 9.3.2014 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  Talk: *Analysis of single and multi phase flows in porous media (I)*

- 6.26.2014 Analysis & PDEs Seminar
  Warwick Mathematics Institute, University of Warwick
  Talk: *Single and multi phase Forchheimer flows in porous media*

- 4.29.2014 Bio-Math Seminars
  Department of Mathematics and Statistics, Texas Tech University
  Talk: *Global solutions of the Shigesada-Kawasaki-Teramoto system*

- 10.1.2013 Colloquium
  Department of Mathematics and Statistics, Texas Tech University
  Talk: *Non-linear Problems in Fluid Dynamics*

- 9.4.2013 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  Talk: *L-infinity estimates for generalized Forchheimer flows*

- 3.20.2013 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  Talk: *Generalized Forchheimer equations for porous media: Part V*

- 4.11.2012 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  Talk: *An interface boundary value problem for incompressible fluids in two-layer domains (II)*

- 4.4.2012 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  Talk: *An interface boundary value problem for incompressible fluids in two-layer domains (I)*

- 10.14.2011 Colloquium
  Department of Mathematics, University of Tennessee, Knoxville
  Talk: *Navier–Stokes equations and geophysical fluid dynamics*

- 9.7.2011 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  Talk: *Forchheimer equations in porous media - Part IV (lecture 2)*

- 8.31.2011 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  Talk: *Forchheimer equations in porous media - Part IV (lecture 1)*

- 4.13.2011 CAMP/Nonlinear PDEs Seminar
  Department of Mathematics, University of Chicago
  Invited Talk: *An explicit Poincaré–Dulac normal form for Navier–Stokes equations*

- 2.9.2011 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  Talk: *An explicit Poincaré–Dulac normal form for Navier–Stokes equations*

- 9.27.2010 PDE/Applied Math Seminar
  Department of Mathematics, Indiana University
  INVITED TALK: *An explicit Poincaré–Dulac normal form for Navier–Stokes equations*

- 9.22.2010 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  TALK: *Forchheimer equations in porous media - Part III (lecture 2)*

- 9.15.2010 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  TALK: *Forchheimer equations in porous media - Part III (lecture 1)*

- 9.9.2009 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  TALK: *Forchheimer equations in porous media - Part II*

- 3.25.2009 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  TALK: *Forchheimer equations in porous media - Part I*

- 9.24.2008 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  TALK: *Navier-Stokes equations in thin domains with Navier friction boundary conditions. Part II*

- 9.17.2008 Applied Mathematics Seminars
  Department of Mathematics and Statistics, Texas Tech University
  TALK: *Navier-Stokes equations in thin domains with Navier friction boundary conditions. Part I*,

- 2.11.2008 Colloquium
  School of Mathematical and Statistical Sciences, Arizona State University
  TALK: *Some problems in geophysical fluid dynamics*,

- 2.5.2008 Colloquium
  Department of Mathematics and Statistics, Texas Tech University
  TALK: *Some problems in geophysical fluid dynamics*,

- 10.17.2007 Institute Seminar
  Department of Mathematics, Indiana University
  INVITED TALK: *Global strong solutions of equations in geophysical fluid dynamics*

- 4.9.2007 PDE and Dynamical Systems Seminars
  School of Mathematics, University of Minnesota
  TALK: *Navier–Stokes equations with Navier boundary conditions in nearly flat domains*

- 4.4.2007 PDE Seminar
  School of Mathematics, University of Minnesota
  TALK: *The normal form of the Navier–Stokes equations in suitable normed spaces*

- 11.27.2006 PDE and Dynamical Systems Seminars
  School of Mathematics, University of Minnesota
  TALK: *On the Stokes and Laplacian operators in Navier–Stokes equations*

- 12.8.2005 Colloquium
  Department of Mathematics, University of Nevada, Reno
  INVITED TALK: *Asymptotic behavior of statistical solutions to the Navier–Stokes equations*

- 10.26.2005 PDE Seminar
  School of Mathematics, University of Minnesota
  TALK: *Normalization maps and statistical solutions in Navier–Stokes equations*

- 10.3.2005 Dynamical Systems Seminar
  School of Mathematics, University of Minnesota
  TALK: *A normal form for the Navier–Stokes equations*

- 4.27.2005 PDE and Dynamical System Seminars
  School of Mathematics, University of Minnesota
  INVITED TALK: *Asymptotic analysis of the helicity in 3D periodic Navier–Stokes equations*

- 2.14.2005 Applied Mathematics Seminar
  Department of Mathematics, Texas A&M University
  TALK: *On the solutions to the normal form of the Navier–Stokes Equations*

- 3.2004 Applied Mathematics Seminar
  Department of Mathematics, Texas A&M University
  TALK: *On the helicity in 3D Navier–Stokes equations*

Organizing Work

- 7.2024 Co-organizer of minisymposium (3 sessions) *Recent developments in the study of fluids (MS53, MS61, MS69)*
  2024 SIAM Annual Meeting
  Spokane, Washington July 8-12, 2024

- 7.2016 Co-organizer of special session (SS108) *New Developments in Porous Media*
  The 11th AIMS Conference on Dynamical Systems, Differential Equations and Applications
  Orlando, Florida, July 1-5, 2016

- 12.2015 Co-organizer of Mini-symposium *Dynamics of Partial Differential Equations*
  SIAM Conference on Analysis of Partial Differential Equations
  Scottsdale, Arizona, December 7-10, 2015

- 4.2014 Co-organizer of Special Session: *Navier-Stokes Equations and Fluid Dynamics*
  AMS 2014 Spring Central Sectional Meeting
  Texas Tech University, Lubbock, TX, April 11-13, 2014

- 12.2013 Co-organizer of Mini-Symposium: *Elliptic and Parabolic Equations with Nonstandard Nonlinearity*
  SIAM conference on Analysis of Partial Differential Equations
  Lake Buena Vista, Florida, December 7-10, 2013

- 10.2013 Co-organizer of *The 13th Red Raider Mini-Symposium*, Department of Mathematics and Statistics, Texas Tech University, October 25-26, 2013

- 6.2013 — Co-organizer of Mini-Symposium: *Dynamics of Non-linear Flows in Porous Media: Analysis and Applications*
  SIAM Conference on Mathematical and Computational Issues in the Geosciences
  University of Padova, Italy, June 17-20, 2013

- 11.2011 — Co-organizer of Mini-Symposium: *Partial Differential Equations for Non-linear Processes in Porous Media*
  SIAM Conference on Analysis of Partial Differential Equations
  San Diego, CA, 11.November 14-17, 2011

- 10.2009 — Co-organizer of *The 9th Red Raider Mini-Symposium*, Department of Mathematics and Statistics, Texas Tech University, October 29-31, 2009

Conferences and Workshops Attended

- 4.2023 — Workshop: Degeneracy of Algebraic Points (Diophantine Geometry Program)
  Mathematical Sciences Research Institute (MSRI)
  Berkeley, CA, April 24–28, 2023

- 10.2009 — AMS 2009 Fall Central Section Meeting
  Waco, TX, October 16-18, 2009

- 7.2009 — Summer Program: Nonlinear Conservation Laws and Applications
  Institute for Mathematics and Applications
  Minneapolis, MN, July 13-31, 2009

Visits

- 3.2023 — Department of Mathematics, Indiana University
- 6.2014 — Mathematics Institue, Uninersity of Warwick

Editorialship

- 5.2022 – present — Member of the editorial board of Sakarya University Journal of Science (SAUJS)

Teaching Experience

- Texas Tech University, Lubbock, Texas

Spring 2026.	MATH3354-001. Differential Equations I
Spring 2026.	MATH4354-001. Differential Equations II
Fall 2025.	MATH3354-002. Differential Equations I
Fall 2025.	MATH3350-122. Higher Mathematics for Engineers and Scientists I
Spring 2025	MATH3351-001. Higher Mathematics for Engineers and Scientists II
Spring 2025	MATH3354-001. Differential Equations I
Fall 2024	MATH4000-004 (4351). Advanced Calculus II
Fall 2024	MATH3351-001. Higher Mathematics for Engineers and Scientists II
Spring 2024	MATH5099-007. Partial Differential Equations II
Spring 2024	MATH4354-001. Differential equations II
Fall 2023	MATH5332-001. Partial Differential Equations I
Fall 2023	MATH4354-001. Differential equations II
Fall 2022	MATH 5099-007. Methods in Partial Differential Equation
Spring 2022	MATH3350-012. Higher Mathematics for Engineers and Scientists I
Spring 2022	MATH3350-111. Higher Mathematics for Engineers and Scientists I
Fall 2021	MATH2450-011. Calculus III with Applications

Fall 2021	MATH2450-013. Calculus III with Applications
Spring 2021	MATH2450-002. Calculus III with Applications
Spring 2021	MATH2450-D01. Calculus III with Applications
Fall 2020	MATH3351-001. Higher Mathematics for Engineers and Scientists II
Fall 2020	MATH2450-011. Calculus III with Applications
Spring 2020	MATH3350-022. Higher Mathematics for Engineers and Scientists I
Spring 2020	MATH3350-021. Higher Mathematics for Engineers and Scientists I
Fall 2019	MATH4351-001. Advanced Calculus II
Fall 2019	MATH2450-022. Calculus III With Applications
Spring 2019	MATH5332-001. Partial Differential Equations I.
Spring 2019	MATH4354-002. Differential equations II
Fall 2018	MATH3354-002. Differential equations I
Fall 2018	MATH3351-002. Higher Mathematics for Engineers and Scientists II
Spring 2018	MATH3354-001. Differential equations I
Spring 2018	MATH4354-002. Differential equations II
Spring 2017	MATH3350-013. Higher Mathematics for Engineers and Scientists I
Spring 2017	MATH3351-001. Higher Mathematics for Engineers and Scientists II
Summer II 2016	MATH3350-202. Higher Mathematics for Engineers and Scientists I
Spring 2016	MATH3351-002. Higher Mathematics for Engineers and Scientists II
Spring 2016	MATH5332-001. Partial Differential Equations I
Fall 2015	MATH3351-004. Higher Mathematics for Engineers and Scientists II
Fall 2015	MATH3350-016. Higher Mathematics for Engineers and Scientists I
Spring 2015	MATH5332-001. Partial Differential Equations I
Spring 2015	MATH4351-001. Advanced Calculus II
Fall 2014	MATH4350-002. Advanced Calculus I
Fall 2014	MATH1320-030. College Algebra
Spring 2014	MATH3310-002. Introduction to Mathematical Reasoning and Proof
Spring 2014	MATH2360-008. Linear Algebra
Fall 2013	MATH2360-001. Linear Algebra
Fall 2013	MATH1451-H01. Calculus I With Applications-Honors
Spring 2013	MATH 5099-011. Partial Differential Equations III
Spring 2013	MATH4354. Differential Equations II
Fall 2012	MATH5333. Partial Differential Equations II
Fall 2012	MATH2450. Calculus III with Applications
Spring 2012	MATH5332. Partial Differential Equations I
Spring 2012	MATH3351. Higher Mathematics for Engineers and Scientists II
Fall 2011	MATH3350-010. Higher Mathematics for Engineers and Scientists I
Fall 2011	MATH3350-012. Higher Mathematics for Engineers and Scientists I
Spring 2011	MATH5341. Functional Analysis II. Section 001
Spring 2011	MATH5332. Partial Differential Equations
Fall 2010	MATH5340. Functional Analysis I
Fall 2010	MATH2360. Linear Algebra
Spring 2010	MATH4354. Differential Equations II
Spring 2010	MATH1352. Calculus II
Fall 2009	MATH3354. Differential Equations I
Fall 2009	MATH1351. Calculus I
Spring 2009	MATH3350. Higher Mathematics for Engineers and Scientists I
Fall 2008	MATH3350. Higher Mathematics for Engineers and Scientists I

- University of Minnesota, Minneapolis, Minnesota

Spring 2008	MATH 1142 Short Calculus
Spring 2008	MATH 1155 Intensive PreCalculus
Fall 2007	MATH 5535 Dynamical Systems and Chaos
Spring 2007	MATH 1031 College Algebra and Probability
Fall 2006	MATH 4606 Advanced Calculus
Spring 2006	MATH 1151 Pre-Calculus II
Fall 2005	MATH 2263 Multivariable Calculus

- Texas A&M University, College Station, Texas

  Fall 2004 M142 Business Mathematics II
  Summer 2004 M152 Engineering Mathematics II (assistant)

- Indiana University, Bloomington, Indiana

  Spring 2002 M025 Pre-Calculus
  Fall 2001 M014 Basic Algebra

- Arizona State University, Tempe, Arizona

  1998–2000 Teaching Assistant
  1999 International Teaching Assistant Seminar

- National University, Hochiminh City

  1997–1998 Instructor
  1996 Teaching Assistant

Students Advised

- 8.2023 Rahnuma Islam, Ph.D. (co-chair of Ph.D. Dissertation Committee)
- 8.2023 Isankaupul Garli Hevage, Ph.D. (co-chair of Ph.D. Dissertation Committee)
- 8.2016 Emine Celik, Ph.D.
- 8.2014 Thinh Kieu, Ph.D. (co-advisor)

Postdoctorates hosted

- 1.2019 – 8.2019 Dr. Phuong Nguyen
- 8.2016 – 8.2019 Dr. Dat Cao

Defense Committee Member

- 6.2023 Mohammad Mahabubur Rahman, Ph.D. Dissertation Committee Member
- 11.2022 Pham Minh Huy Huynh, Ph. D. Defense Committee Member, Universität Klagenfurt, Austria
- 2020 Rahnuma Islam, Master's Thesis Committee Member
- 2019 Thakshila Gunasingha, Master's Thesis Committee Member
- 2014 Anna Krylova, Master's Thesis Committee Member
- 2010 – 2013 Lidia Bloshanskaya, Ph.D. Dissertation Committee Member
- 2012 Pooya Aavani, Master's Thesis Committee Member
- 5.2011 Jedidiah Gohlke, Master's Thesis Committee Member
- 11.2010 Lidia Bloshanskaya, Master's Thesis Committee Member

Service

- University

2010 – present	Member of Graduate Faculty, Graduate School
Fall 2013, 2018, Aug. 2023	Faculty representative at the Graduation Commencement
6.14.2013	Dean's representative, Ph.D.'s defense, David Kimberly, Environmental Toxicology
3.25.2013	Dean's representative, Ph.D.'s defense, Taskin Karim, Chemical Engineering
3.22.2013	Dean's representative, Ph.D.'s defense, Vance Ginn, Economics
6.24.2011	Dean's representative, Ph.D.'s defense, Guangqiu Qin, Environmental Toxicology
6.1.2011	Dean's representative, Ph.D.'s defense, Patrick Mclaurin, Chemistry

- Department

9.2023 – present	Co-organizer of Analysis Seminars
9.2023 – present	State Employee Charitable Campaign (SECC) coordinator for Department of Mathematics and Statistics
1.2017 – 5.2021	Co-organizer of Analysis Seminars (except on-leave Fall 2017)
4.2015 – present	Peer teaching evaluations for fellow faculty
2011 – present	Member of Examination Committee for Preliminary Examination in PDEs
2009 – present	Teaching evaluations for (graduate) teaching assistants
9.2020 – 5.2021	Co-organizer of Pure Mathematics Colloquium: Current Advances in Mathematics
10.2019 – 5.2022	Co-advisor of students' Actuarial Science Group
9.2018 – 12.2020	State Employee Charitable Campaign (SECC) coordinator for Department of Mathematics and Statistics
Fall 2014-Spring 2015	Member of Hiring Subcommittee: Complex Analysis
Spring 2013	Member of Travel Policy Committee
9.2012 – 9.2013	Member of Graduate Committee
12.2009	Substituting member in the Committee, Master's thesis defense, James Woodley, Mathematics
8.2008 – 5.2016	Co-organizer of Applied Mathematics Seminars

- Professional

Expert Reviewer	Ph.D. Dissertation Reviewer for Pham Minh Huy Huynh, Universität Klagenfurt, Austria, 2022

Referee	Acta Applicandae Mathematicae Asymptotic Analysis Bulletin des sciences mathématique Canadian Journal of Mathematics Communications on Pure and Applied Analysis Discrete and Continuous Dynamical Systems Electronic Journal of Differential Equations Hacettepe Journal of Mathematics and Statistics Indiana University Mathematics Journal Journal of Applied Analysis and Computation Journal of Differential Equations Journal of Dynamics and Differential Equations Journal of Mathematical Analysis and Applications Journal of Mathematical Fluid Mechanics Journal of Mathematical Physics Multiscale Modeling and Simulation (SIAM Interdisciplinary Journal) Nonlinearity Nonlinear Analysis Series A: Theory, Methods and Applications Numerical Methods for Partial Differential Equations Qualitative Theory of Dynamical Systems Studies in Applied Mathematics

Awards

- 2005 *L. F. Guseman Award,* Texas A&M University
- 2002 – 2005 *Departmental Graduate Fellowship,* Texas A&M University
- 2002 *AUF Fellowship,* Texas A&M University
- 2001 *James P. Williams Memorial Award,* Indiana University
- 2001 *Eberhard E. Hopf Fellowship,* Indiana University
- 1997 *Outstanding Student Award,* National University, Hochiminh City

Other information

- Citizenship U. S. A.
- Languages English, Vietnamese